

PROJECT 2 INSTRUCTIONS

Please also pay close attention to any additional specifications provided by your professor. Professors often will clarify their expectations regarding the format and presentation of your submission.

Topic – Causes of Death in 1980 and 2016

According to the 1980 Census, the United States population was approximately 226,540,000 in 1980. It grew to approximately 323,120,000 at the beginning of 2016. Using Census data for 1980 and estimates derived from mortality data for 2016, we arrive at the population estimates given in the table below:

Year	Total Population	Ages 15–24	Ages 25–44	Ages 45–64
1980	226,540,000	42,475,000	62,707,000	44,497,000
2016	323,120,000	43,500,000	85,150,000	84,300,000

The National Center for Health Statistics published a document entitled “Health, United States, 2015: With Special Feature on Racial and Ethnic Health Disparities” that includes a table listing the leading causes of death in 1980 by age bracket. The CDC further produced a National Vital Statistics Reports that provided similar information for the year 2016. Under the Project Instructions link in Canvas, you’ll find **Project 2: Data**, a spreadsheet showing the leading causes of death in both 1980 and 2016 for these 3 age categories. Each of the first 3 questions has both a computational part and a discussion part. To get full credit for each of the discussion parts, please cite a reference to support your claims. This should not be an excessively difficult task: you can easily find information online for most of the illnesses or other causes listed in the report. All you need to do is provide the website you used (though other resources are also permitted if you prefer to use one of those).

1. Assuming that the population numbers in the above table are relatively accurate, use the Project 2: Data spreadsheet to compute the deaths per 1000 people for each age group in both 1980 and 2016. Deaths per 1000 people is computed using the formula Deaths per 1000 = (# of deaths) / (total population) × 1000. Do not round your answer to the nearest whole number, provide at least 2 decimal places (but no more than 4). Give these 6 values (e.g. deaths per 1000 people for ages 15–24 in 1980) and then cite a reference to discuss what might account for the changes between the deaths per 1000 in 1 of these 3 age categories between 1980 and 2016. Your discussion should be at least 50 words.
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$$\text{Deaths per 1000} = (\text{Number of Deaths} / \text{Total Population}) \times 1000$$

Let's compute the deaths per 1000 for each age group in both 1980 and 2016:

Ages 15–24

- **1980:**
 - Total Deaths = 49,027
 - Total Population = 42,475,000

$$\text{Deaths per 1000} = (49,027 / 42,475,000) \times 1000 \approx 1.15$$

- **2016:**
 - Total Deaths = 32,575
 - Total Population = 43,500,000

$$\text{Deaths per 1000} = (32,575 / 43,500,000) \times 1000 \approx 0.75$$

Ages 25–44

- **1980:**
 - Total Deaths = 108,658
 - Total Population = 62,707,000

$$\text{Deaths per 1000} = (108,658 / 62,707,000) \times 1000 \approx 1.73$$

- **2016:**
 - Total Deaths = 135,408
 - Total Population = 85,150,000

$$\text{Deaths per 1000} = (135,408 / 85,150,000) \times 1000 \approx 1.59$$

Ages 45–64

- **1980:**
 - Total Deaths = 425,338
 - Total Population = 44,497,000

$$\text{Deaths per 1000} = (425,338 / 44,497,000) \times 1000 \approx 9.56$$

- **2016:**
 - Total Deaths = 539,961
 - Total Population = 84,300,000

$$\text{Deaths per 1000} = (539,961 / 84,300,000) \times 1000 \approx 6.41$$

Summary of Deaths per 1000 People (1980 vs. 2016):

- **Ages 15–24:**
 - 1980: 1.15 deaths per 1000
 - 2016: 0.75 deaths per 1000
- **Ages 25–44:**
 - 1980: 1.73 deaths per 1000
 - 2016: 1.59 deaths per 1000
- **Ages 45–64:**
 - 1980: 9.56 deaths per 1000
 - 2016: 6.41 deaths per 1000

2. Besides the changes in the overall death rate in the past 3 decades, the leading causes of death vary somewhat between 1980 and 2016. Choose 1 of the 3 age ranges and select 1 cause of death from the Project 2: Data spreadsheet that strikes you as noteworthy and that appears in both the 1980 and 2016 lists. For the cause of death that you selected, compute the number of deaths per 1000 in both 1980 and 2016 for your chosen age group. Do not round your answer to the nearest whole number, provide at least 2 decimal places. Cite a reference to discuss the possible reasons for any changes in the rates over this period. Your discussion should be at least 50 words.

Ages 15-24: Unintentional Injuries

- 1980:
 - Total Deaths from Unintentional Injuries = 26,206
 - Total Population = 42,475,000

$$\text{Deaths per 1000} = (26,206 / 42,475,000) \times 1000 \approx 0.62$$

- 2016:
 - Total Deaths from Unintentional Injuries = 13,895
 - Total Population = 43,500,000

$$\text{Deaths per 1000} = (13,895 / 43,500,000) \times 1000 \approx 0.32$$

Discussion:

3. The decrease in the number of deaths from unintentional injuries in the Ages 15-24 group could be attributed to improvements in vehicle safety, public health campaigns, and advancements in emergency medical care. For example, seat belt use and road safety education have reduced the rates of car accidents in younger populations.

Reference:

For a reference, you can look at safety improvements and injury prevention measures from sources like the Centers for Disease Control and Prevention (CDC) website or other trusted health organizations.

Not only do the leading causes of death vary across time, they vary significantly for different age ranges. Looking only at the 2016 data, choose a cause of death that appears in both the 25–44 and 45–64 age categories and compute the number of deaths per 1000 people for both age categories. Do not round your answer to the nearest whole number, provide at least 2 decimal places. Cite a reference to discuss a possible reason for any differences in these values as people advance in age. Your discussion should be at least 40 words. Diseases of Heart in 2016:

- Ages 25-44:
 - Deaths from Diseases of Heart = 13,922
 - Total Population = 85,150,000

$$\text{Deaths per 1000} = \frac{13,922}{85,150,000} \times 1000 \approx 0.16$$

- Ages 45-64:
 - Deaths from Diseases of Heart = 112,637
 - Total Population = 84,300,000

$$\text{Deaths per 1000} = \frac{112,637}{84,300,000} \times 1000 \approx 1.34$$

Discussion:

The higher rate of heart disease deaths in the 45-64 age group compared to the 25-44 group is expected. As people age, the risk of heart disease increases due to factors like hypertension, high cholesterol, and lifestyle choices. The 25-44 age group is less likely to suffer from heart disease as it typically affects older individuals.

Reference:

A good reference to explore this could be from the American Heart Association (AHA) or the CDC on heart disease risk factors.

4. Contemplating causes of death might strike some people as unpleasant or even morbid. However, the Bible encourages us to give some thought to the fact of our own mortality. Ecclesiastes 7:2–4 says: “It is better to go to the house of mourning than to go to the house of feasting, for this is the end of all mankind, and the living will lay it to heart. Sorrow is better than laughter, for by sadness of face the heart is made glad. The heart of the wise is in the house of mourning, but the heart of fools is in the house of mirth.” It is interesting to consider why the author of Ecclesiastes encourages the wise to go to the house of mourning and the living to lay the end of all mankind to heart. What value might there be in thinking about the, admittedly uncomfortable, subject of the end of all mankind? What comes to your mind when you consider this topic? Please respond with at least 100 words.

The project involves computing death rates and analyzing mortality data over time. For the computational part, you’ve calculated the deaths per 1000 people for three age groups, as well as comparing different causes of death. The reflections in the discussion sections provide context to these numbers, explaining the trends and potential reasons behind the changes in mortality rates.

Submit your Project by 11:59 p.m. (ET) on Monday of Module/Week 5.